

Report on 4-Level Elevator Control System

1. Objective of the Circuit

This circuit is designed to simulate an elevator that can go to four floors: **0, 1, 2, and 3**. It lets the user select a floor, and the circuit will remember the selected floor using memory (flip-flops). An LED will light up to show which floor has been selected.

2. Main Components Used

a. Floor Buttons (F0–F3)

- These are input buttons to choose a floor.
- You press **one button** to select a floor.
- Only one floor should be selected at a time.

b. Enable (EN)

- A control input that allows the circuit to **accept a floor button press**.
- If $EN = 1$, the circuit will load the selected floor.
- If $EN = 0$, the circuit will ignore button presses.

c. Reset

- Clears all floor selections.
- When active, it turns **all LEDs off** (no floor is selected).

d. Clock

- A timing signal that tells the circuit when to check the inputs.
- On each **rising edge** (0 to 1), the circuit updates its memory.

e. D Flip-Flops (4)

- These are memory units that **store the selected floor**.
- One D flip-flop is used for each floor.
- Each one saves a '1' if its floor is selected, otherwise stores '0'.

f. LEDs

- Each flip-flop is connected to an LED.

- If a flip-flop stores '1', the LED for that floor turns **on**.
 - It shows which floor is currently selected.
-

3. How the Circuit Works (Step by Step)

Step 1: Reset the Elevator

- If Reset = 1 and the clock ticks, **all floor selections are cleared**.
- All flip-flops store 0, so **all LEDs turn off**.

Step 2: Choose a Floor

- Press one floor button (like F2).
- Make sure **EN = 1**, and wait for the **next clock pulse**.
- The circuit will now **store the value** in the D input of flip-flops.

Step 3: Update Memory

- On the clock's rising edge:
 - The selected floor's flip-flop gets 1.
 - All other flip-flops get 0.
 - Example: If F2 is pressed, Q2 = 1, and Q0, Q1, Q3 = 0.

Step 4: Show the Selected Floor

- The flip-flop outputs (Q0–Q3) go to LEDs.
- So if Q2 = 1, **LED2 turns ON**, showing floor 2 is selected.

Step 5: Hold the Selection

- After selecting a floor, set **EN = 0**.
 - Now the flip-flops hold their value even if buttons are pressed.
-

4. Conclusion

This 4-floor elevator circuit uses **flip-flops, floor buttons, and LEDs** to simulate an elevator's memory and response system. It uses a **clock** for proper timing, an **enable** to load only when needed, and a **reset** to clear everything. The design is simple, clear, and effective for learning basic digital circuit concepts.